IOP Conference Series:
Earth and
Environmental Science

PURPOSE-LED
PUBLISHING™

PAPER • **OPEN ACCESS**

Application of biostimulant consortium to increase the growth of sugarcane (Var. Cenning) in a dry land

To cite this article: S Wahyuni *et al* 2024 *IOP Conf. Ser.: Earth Environ. Sci.* **1306** 012001

View the article online for updates and enhancements.

You may also like

- Effect of liquid seaweed extracts as biostimulant on vegetative growth of soybean
Z A Noli, Suwirmen, Aisyah et al.

- Effect of biostimulant and silica application on sugarcane (*Saccharum officinarum* L.) production
L W Anggraeni, A F Pratama, P H Putri et al.

- The effect of biostimulant in root and population of phosphate solubilizing bacteria: a study case in upland rice
Agustin Sri Mulyatni, R H Praptana and D Santoso

This content was downloaded from IP address 200.130.54.87 on 06/02/2026 at 14:56

Application of biostimulant consortium to increase the growth of sugarcane (Var. Cenning) in a dry land

S Wahyuni[1], D N Kalbuadi[1], M E R B Prasetyo[1], S M Putra[1], H Widiastuti[1] and Priyono[1]

[1] Indonesian Research Institute for Biotechnology and Bioindustry, Jl. Taman Kencana No.1, Bogor, Indonesia

Corresponding author email: sri09wahyuni@gmail.com

Abstract. Sugarcane is considered one of the most promising crops. The growth of the crop is affected by photosynthesis and toter and nutrient absorption by plants. Biostimulant is a non-nutrient biomaterial that is proven to improve crop growth. Based on this idea, Biostimulant technology can be used as an alternative to enhance sugarcane growth. This research aimed to assess the Biostimulant consortium (Sucrosin, humic acid, and AM fungal) to improve the growth of the sugarcane var. Cenning in a dry land. The study tested two treatments, i.e., control (standard procedures) and Biostimulant Consortium application. Sucrosin was applied on 1, 3, 4, and 5 months after planting (MAP) by using a foliar spray. Application of mycorrhizal fungi was spread around the roots area of sugarcane crops. In contrast, the application of humic acid was sprayed on the ground as a coating agent for inorganic fertilizers. Coating fertilizer is carried out by mixing a humic acid solution with as much as 1% of the fertilizer. The research showed the treatment of the Biostimulant Consortium yielded significant results in the sugarcane growth compared to the control on 8 MAP. Biostimulant Consortium can improve crop height and stem diameter significantly up to 30-40% compared to control. It indicates that the Biostimulant consortium's treatment can improve sugarcane's vegetative growth in s dry land.

Keywords: sugarcane, biostimulant consortium, Cenning, non-nutrient biomaterial

1.Introduction

Globally, Indonesia is the largest sugar-importing country. Based on the Ministry of Trade data, the Indonesian raw sugar import allowance is increasing yearly, from 3.1 million tonnes in 2014 to 3.6 million tonnes in 2015. Therefore, the effort to increase national sugar production through extensibility has led to the utilization of marginal land, including dry land [1]. The area of dry land for sugarcane currently reaches 171 thousand hectares and is projected to increase to 260 thousand hectares in the year 2050 [2]. However, the dry land limits sugarcane growth because of drought stress. Sugarcane crops undergoing drought stress result in suboptimal shoots and other vegetative growth, decreasing sugarcane production per hectare [1]. Sugarcane crops in drought stress will cause the shoots and vegetative growth to be sub-optimal and reduce sugarcane production per hectare.

The current effort to improve sugarcane productivity is still focused on fertilizing technology. Therefore, biostimulant technology can be used as an alternative to the present technology. Biostimulant is a small number of organic compounds that can support and improve plant growth. It improves nutrient absorption efficiency and tolerance of abiotic, thus improving production[3].

Content from this work may be used under the terms of the Creative Commons Attribution 3.0 licence. Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.
Published under licence by IOP Publishing Ltd

Biostimulants increased the growth of sugarcane crops, horticulture, and some food crops [4, 5, 6, 7]. The Biostimulant in this research is made from extracts of seaweed organic matter. Applying organic biostimulants from seaweed has considerably influenced the growth of sugarcane crops. The use of biostimulants with humic acid and Mycorrhiza increased the vegetative growth of sugarcane, such as height, stem diameter, and plant rooting [8].

Humic acid and Myoriza can increase plant growth. The negative charge in humic acid makes it effective as a macronutrient, micronutrient, and water molecule and also can reduce leaching. It can increase the availability and intake of nutrients for plants [9]. Roni and Soedarmaji [10] reported that the administration of humic acid improved the growth and production of tropical kudzu crops. The same research was delivered by [9]; the application of humic acid on corn plants in dry land proved to increase plant growth (weight, height, and nutritional content of fruit). At the same time, Mycorrhiza has been widely reported to be a biological agent functioning to increase the availability of nutrients in soil [11].

Varieties of Cenning used in this research are based on the Decree Release 3679/Kpts/SR. 120/11/2010. This variety has potential production of 775 kg/ha with a yield of 10.97%. Cenning has agronomic properties of having the diameter of a medium bar density, suitably developed on the type of alluvial and Mediterranean soil with a very high clay content with enough watering/rainfall. This study aims to justify Biostimulant Consortium Technology (BCT) to increase the vegetative growth of sugarcane crops in a dry land.

2. Materials and Methods

This field experiment was conducted in Wanayasa, Tegal Regency, Central Java. This activity was conducted in January-September 2019 using the sugar cane varieties Cenning. The BCT consists of Sucrosin, Humacoat (humic acid), and MizaPlus (Mycorrhiza), which is a product of the Indonesian Research Institute of Biotechnology and Bioindustry (IRIBB).

This experiment was conducted by comparing control and BCT treatment plots. Each plot consists of 5 observation plates of 15 meters in length on a 4 ha land area. Each treatment was performed at different times. Sucrosin was applied on 1, 3, 4, and 5 MAP (a month after planting) with foliar spray. The dose of Sucrosin amounted to 375 ml/ha at 3 MAP and 250 ml/ha at 1, 4, and 5 MAP. MizaPlus was applied by spreading around the plant at a 60 kg/ha dose. Humacoat was applied in 2 stages: spray on the soil and coating of inorganic fertilizer. The application of Humacoat spray on the ground is carried out by using the foliar spray at 1 MAP. The Humacoat dose is 6 liters/ha + 200 liters of water. Fertilizer coating was done by mixing inorganic fertilizer with 1% solution Humacoat (Humacoat + urea, 3:2). Fertilizer dose given in plants as much as 800 kg Za + 200 kg SP36 + 200 kg KCL. The observations were carried out at 2, 3, 4, 5, 6, and 8 MAP. The observed parameters consist of the number of plants, the height of the plant, the diameter of the stems, and the number of stem sections. Data on the observation results are analyzed statistically using the Independent sample T-Test.

3. Results and Discussions

The experimental parameters (vegetative plant) include the number of plants, the plant's height, the stem's diameter, and the number of branches. The observations during 2-6 MAP show that the BCT treatment significantly improves sugarcane growth compared to the control, especially in the number of plants per meter, plant height, and trunk diameter.

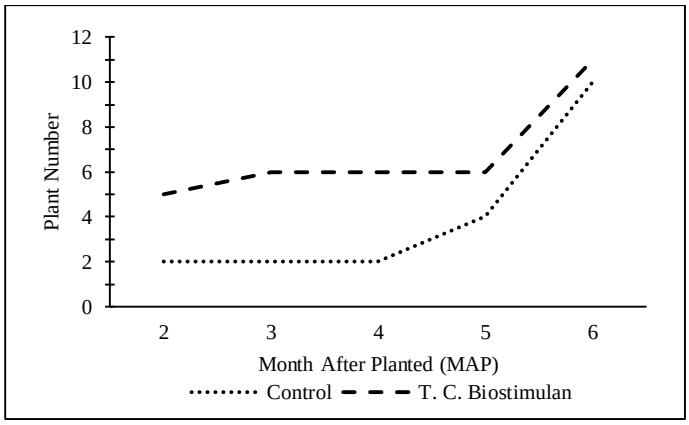

Figure 1. The number of plants per meter of sugarcane on the 2-6 MAP.

The number of plants per meter of sugarcane plants at 2-6 MAP showed in Figure 1. The BCT treatment has better results ranging from 2 MAP. It indicates that the treatment of a bio-stimulatory consortium can support the inadequate and multiply the number of plants per meter. The role of BCT treatment has a significant influence at 3-4 MAP with an increase of control of 200%. At 4-6 MAP, the number of plants per meter in sugarcane plants with a BCT treatment has decreased and increased by 20-50% compared to the control. This result may be due to the influence of drought conditions in the experimental land that began when 4 MAP until the end of observation.

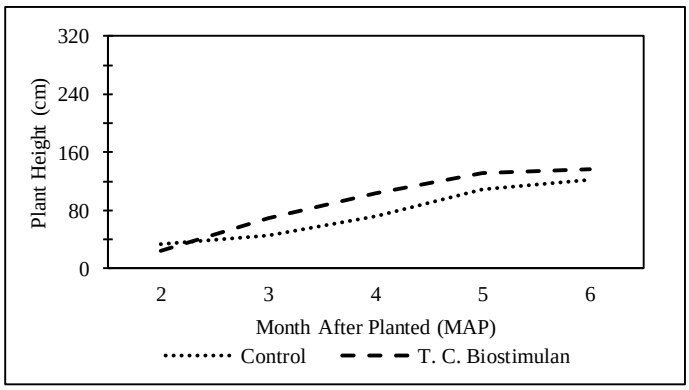

Figure 2. High observation of sugarcane on the 2-6 MAP.

High Observation of sugarcane crops at the age of 2-6 MAP (Figure 2) showed the treatment of Biostimulant consortium tends to improve the plant height compared to the control. The time of crop two that MAP sugar cane plant is still higher compared to the treatment of a bio-stimulatory consortium. However, when the plant entered the age of 3, high MAP sugarcane plants with a consortium treatment increased by 51% compared to the control. So sugarcane plants with a consortium treatment can be higher than control up to 6 MAP.

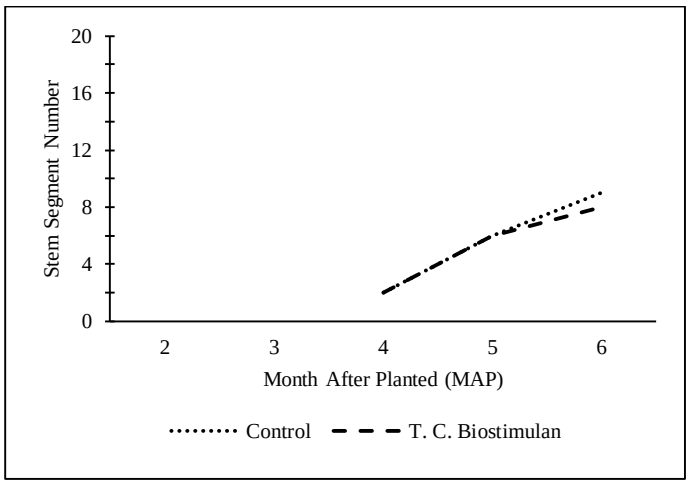

Figure 3. A steam segment of sugarcane on the 4-6 MAP.

Stem section numbers observation of sugarcane was newly performed when 4-6 MAP (Figure 3). The sugarcane plant in this experiment began to emerge when plants aged 4 MAP. Figure 3 shows that the BCT treatment has not significantly influenced the addition of the number of sections in the trunk. However, the number of sections on the stem of the control sugarcane plants on the 6 MAP was higher than in the BCT treatment. The time when 4-6 of the land's MAP has a drought to interfere with the growth of sugarcane plants might cause this condition.

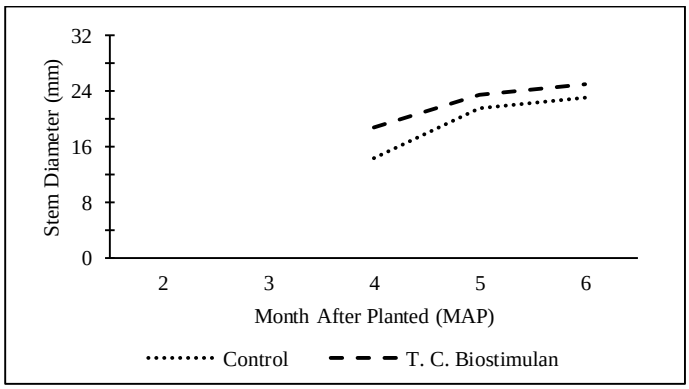

Figure 4. Steam diameter of sugarcane on the 4-6 MAP.

The diameter of the stem in 4-6 MAP (Figure 4) on the BCT treatment also shows better than the control plants. Stem Diameter can be measured when sugarcane plants have begun to appear on the stem. The sugarcane plant in this experiment began to emerge in 4 MAP. The BCT treatment in 4 MAP can produce a larger diameter than the control, with an increase of 31%. The diameter of the trunk in 5-6 MAP at BCT treatment is still higher than the control plants, but the value of growth was decreased at 4 MAP. A shortage of water due to the drought that influences stem diameter growth may be one of the causes.

The decrease of sugarcane growth in this experiment is both on control and BCT treatment in 4-6 MAP was caused by the drought factor. According to Irianto [1], sugarcane plants in dry land or under drought stress will undergo stunted initial vegetative growth. This initial growth greatly determines further growth and development, including productivity. Generally, sugarcane plants can grow normally when the vegetative growth period (until 6 MAP) gets enough water. However, the BCT treatment can influence the control of observed parameters. It can be the better initial capital plant for the growth and development of advanced sugarcane plants.

The final vegetative growth (8 MAP) demonstrates the influence of a significant BCT treatment compared to the control in the number of plants, the diameter of the trunk, and the number of plant sections (Table 1).

Table 1. The final vegetative growth of sugarcane on 8 MAP.

Tsugarcanes	Plant Number		Plant Height		Stem Diameter		Stem Segment Number	
	plant	delta, %	cm	delta, %	mm	delta, %	segment	delta, %
Control	8,99*	0,00	111,6ns	0,00	22,2*	0,00	11**	0,00
T. C. Biostimulan	11,8*	31.31	137,8ns	9,01	24,22*	23,48	15**	39,62

The significant result indicates that the application of the biostimulant consortium is still able to maintain the growth of sugarcane plants compared to the control at 8 MAP when experiencing drought-bleeding conditions. The increase in the parameter height of the plant was 23%, and the trunk diameter was 9% compared to the control. While the number of segments in the stem had significantly increased, that was 40% compared to the control. The provision of biostimulants that can increase vegetative growth was also expected to improve the harvest of sugarcane crops [4].

Biostimulants that contain a complex mixture of polysaccharides, micronutrients, and regulatory hormones, can positively influence the process of photosynthesis, cell metabolism, nitrogen, sulfur metabolism, and stress drought response [6]. Wahyuni et al. [8] reported that a complete biostimulant (phytohormone, humic acid, and Mycorrhiza) could provide better vegetative growth. The application of biostimulants in agricultural activities has a positive impact on plants. The biostimulant application in plants can improve seed germination rates, enhance the mobilization and distribution of nutrients, improve rooting, flowering, fruit, and crop yields, and increase the content of chlorophyll and area of the leaves, as well as increase resistance to biotic and abiotic stress [12].

The application of Humacoat for increasing plant growth includes soil improvement and fertilizer coating. The ability of Humacoat as groundwater is related to the humic acid fraction as a provider of nutrients such as N, P, K, and S into the soil, and also C elements that act as an energy source for microbes. Humacoat ability as a chelator and coating of inorganic fertilizer is related to the content of a negatively charged functional group derived from the H ion dissociation that causes humic acid to have a high ion exchange capacity to bind inorganic fertilizer [13]. Alimin [14] reported that additional humic acid as polyelectrolytic macromolecular material with its functional group (-COOH and-OH phenolate and-OH Alcoholic) could provide a complex function with metal ions due to deprotonation at a relatively high pH. It can support humic acid-forming to a complex bond with ions released by inorganic fertilizer (N, P, K), which triggers slow-release fertilizer to be more stable.

The application of MizaPlus plays a role in increasing sugarcane growth in the land because it can improve the rooting condition by the additional Arbuscular Mycorrhiza. Arbuscular Mycorrhiza plays a role in enhancing the root's ability to absorb nutrients from the soil [15]. In addition, Ruiz-Lozano and Azcón [16] have also reported that Arbuscular Mycorrhiza could increase crop resistance in drought stress through increased nutrient absorption, transpiration of leaves, and efficiency of water use.

4. Conclusions

Treatment of the biostimulant consortium (Sucrosin, Humacoat, and MizaPlus) can increase vegetative growths of sugarcane on varieties Cenning for seeds harvest in a dry land, based on parameters trunk diameter, plant height, and sections number with the increase of 9%, 23%, and 40%, respectively against control, at 8 MAP.

5. References

[1] Irianto G 2003 Tebu Lahan Kering Dan Kemandirian Gula Nasional. *Tabloid Sinar Tani* 20 Agustus 2003 [Avaiable at: http://www.litbang.pertanian.go.id/artikel.php/28/]

[2] Mulyani A, Ritung S and Las I 2011 Potensi dan ketersediaan sumberdaya lahan untuk mendukung ketahanan pangan *Jurnal Litbang Pertanian* Vol **30** No 2 pp 73-80
[3] Du Jardin P 2015 Plant Biostimulants: Definition, Concept, Main Categories, and Regulation. *Scientia Horticulturae* Vol **196** pp 3-14.
[4] Putra S M, Santoso D, Pardal S J, Umahati B K, Amanah D M and Susanti P 2017 Pengaruh Biostimulan terhadap pertumbuhan vegetatif tanaman tebu varietas PSJT-941 *Menara Perkebunan* Vol **85** (1) pp 37-43 DOI: http://dx.doi.org/10.22302/iribb.jur.mp.v85i1.241
[5] Amanah D M and Putra S M 2018 Pengaruh Biostimulan terhadap toleransi kekeringan dan pertumbuhan tanaman tebu varietas Kidang Kencana di Rumah Kaca *Menara Perkebunan* Vol **86** (1) pp 46-55 DOI: http://dx.doi.org/10.22302/iribb.jur.mp.v1i1.287
[6] Calvo P, Nelson L and Kloepper J W 2014 Agricultural uses of plant biostimulants *Plant and Soil* Vol **383 (1) p**p 3-41
[7] Santoso D and Priyono 2014 Proses Produksi dan Formulasi Biostimulan dari Alga Coklat *Sargassum Sp*. serta Penggunaannya untuk Pertumbuhan Tanaman. Indonesia Patent P-00201406718. 2014
[8] Wahyuni S *et al*. 2018 Biostimulasi pertumbuhan vegetatif tanaman tebu (*Saccharum officinarum* L.) pada fase awal di lahan kering *Menara Perkebunan* Vol **86** No 2 pp 91-95 DOI: http://dx.doi.org/10.22302/iribb.jur.mp.v86i2.284
[9] Hermanto D D, Kurnianingsih N K T, Kurnianingsih R and Kamali S R 2013 Pengaruh asam humat sebagai pelengkap pupuk terhadap ketersediaan dan pengambilan nutrien pada tanaman jagung di lahan kering Kec. Bayan-NTB *Jurnal Ilmu Pertanian* Vol **16** No 2 pp 28-41 DOI: https://doi.org/10.22146/ipas.2531
[10] Roni N G K, Soedarmaji and Setiadi Y 2005 Pertumbuhan dan produksi Kudzu Tropika (Pueraria Phaseoloides BENTH.) yang diberi asam humat dan pupuk Fosfat *Majalah Ilmiah Peternakan* Vol 8 (2) pp 1-10
[11] Daras U, Sobari I, Trisilawati O and Towaha J e*t al*. 2015 Effect of mycorrizha and NPKMG fertilizer on growth and production of Arabica coffee *Jurnal Tanaman Industri dan Penyegar* Vol **2** (2) pp 1-8
[12] Sharma H S S e*t al*. 2014 Plant biostimulants: a review on the processing of macroalgae and use of extracts for crop management to reduce abiotic and biotic stresses *J. Appl. Phycol.* Vol. **26** Pp 465-490 DOI: https://doi.org/10.1007/s10811-013-0101-9
[13] Stevenson F J 1994 *Humus Chemistry: Genesis, Composition, Reactions 2nd Edition* (New Jersey: John Wiley & Sons)
[14] Alimin N 2000 Fraksinasi asam humat dan pengaruhnya pada kelarutan ion logam Seng (II), Kadmium (II), Magnesium (II) dan Kalsium (II) [Thesis] FMIPA Universitas Gadjah Mada
[15] Wardhika C, Hadisutrisno B and Widada J 2015 Potensi jamur Mycorrhiza Arbuskular unggul dalam peningkatan pertumbuhan dan kesehatan bibit tebu *(Saccharum Officinarum* L.*) Ilmu Pertanian* Vol **18** (2) pp 84-91
[16] Ruiz-Lozano J M and Azcón R 2000 Symbiotic efficiency and infectivity of an autochthonous arbuscular mycorrhizal *Glomus Sp*. from saline soils and *Glomus deserticola* under salinity *Mycorrhiza* vol **10** no 3 pp 137-143